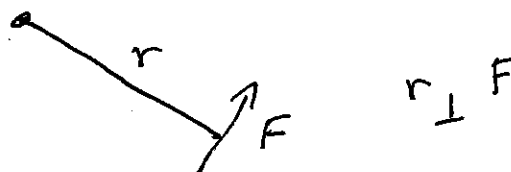
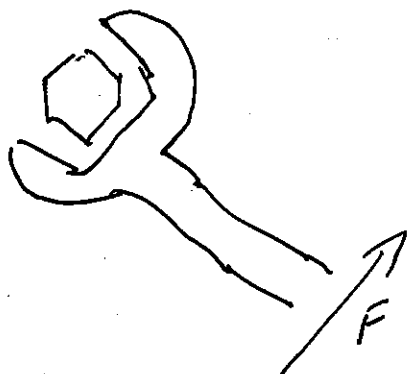


STATICS 12

1/10

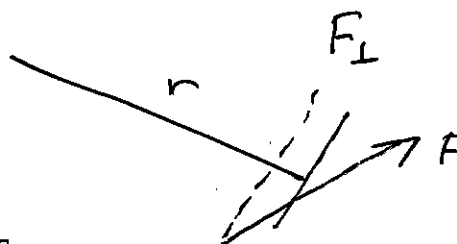
Torque is a twist.

Think wrench or screwdriver



$$\tau = r F$$

$$\tau = r_{\perp} F = r F_{\perp}$$



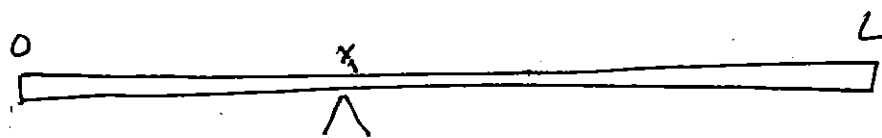
Static means not moving

$$\sum F_x = \sum F_y = \sum F_z = 0$$

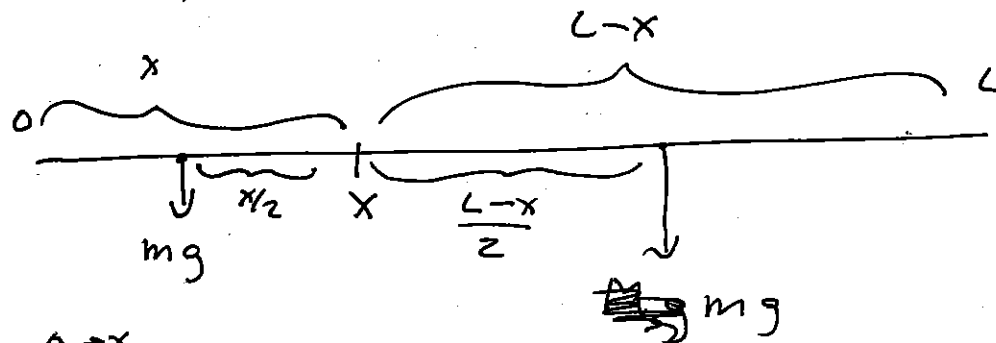
$$\sum \tau_x = \sum \tau_y = \sum \tau_z = 0$$

STATICS 12

21/10



Beam of length L , mass M , where do you balance it?



$$0 \rightarrow x$$
$$m = \frac{x}{L} M$$

$$x \rightarrow L$$
$$m = \frac{L-x}{L} M$$

$$\tau = rF$$

$$\sum \tau_x: \frac{x}{2} \frac{x}{L} M g - \frac{L-x}{2} \frac{L-x}{L} M g = 0$$

$$\frac{x^2}{2L} = \frac{(L-x)^2}{2L}$$

$$x^2 = (L-x)^2$$

$$x = L-x$$

$$L = 2x$$

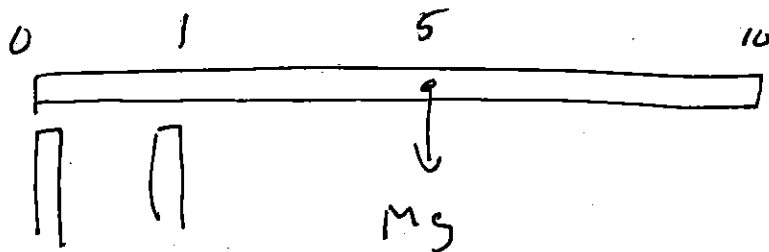
$$x = \frac{L}{2} \quad \text{duh!}$$

DIVING BOARD

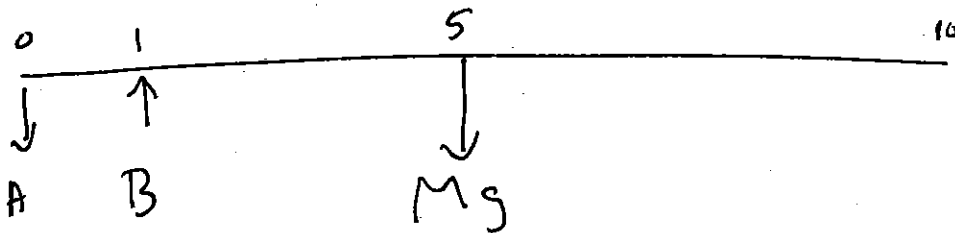
3/10

$$L = 10 \text{ m}$$

$$M = 1000 \text{ kg}$$



A B



$$\Sigma F_y: B - A - Mg = 0 \quad A = ? \quad B = ?$$

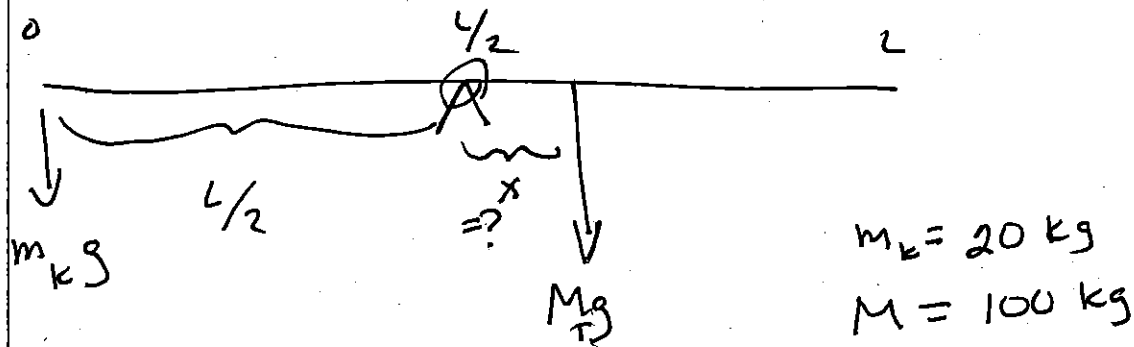
$$\Sigma \tau_A \text{ gives } B \quad A \text{ is eliminated } \tau = r F \quad r = 0$$

$$: 1B - 5Mg = 0 \quad \boxed{B = 5Mg}$$

$$\Sigma F \text{ eq gives } A = B - Mg = 4Mg$$

$$\text{or } \Sigma \tau_B: 1A - 4Mg = 0 \quad \boxed{A = 4Mg}$$

Seesaw length L , mass M (don't care!) **4/10**



$$m_k = 20 \text{ kg}$$

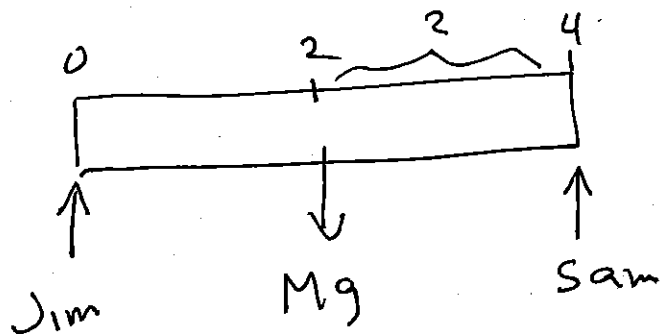
$$M = 100 \text{ kg}$$

$$\sum \tau_{\text{center}}: \frac{L}{2} m_k g - x M_T g = 0$$

$$\frac{L}{2} m_k g = x M_T g$$

$$x = \frac{L m_k}{2 M_T} = \frac{L \cdot 20}{2 \cdot 100} = \boxed{\frac{L}{10}}$$

2 men carry a sofa. Each holds at end



4m long

$m = 50 \text{ kg}$

$$\sum \tau_{\text{Jim}}: -2Mg + 4S = 0$$

$$S = \frac{2Mg}{4} = \frac{1}{2}Mg$$

$$\sum \tau_{\text{Sam}}: 2Mg - 4J = 0$$

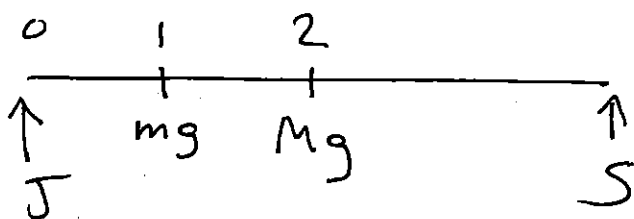
$$J = \frac{1}{2}Mg$$

245N

When you move people their 20kg
kid gets under your feet to 'help'.

5/10

Set them on the sofa 1m from Jim.



$$\sum \tau_{Jim} : -1mg - 2Mg + 4S = 0$$

$$S = \frac{1}{4}(mg + 2Mg) = \frac{1}{4}(20 \cdot 9.8 + 2 \cdot 50 \cdot 9.8)$$

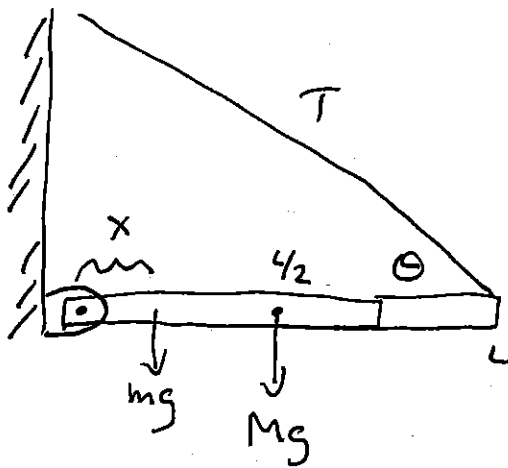
$$= 30 \cdot 9.8 = \boxed{294 \text{ N} = S}$$

$$\sum \tau_{Sam} : -4J + 3mg + 2Mg = 0$$

$$J = \frac{1}{4}(3mg + 2Mg) = \frac{1}{4}(3 \cdot 20 \cdot 9.8 + 2 \cdot 50 \cdot 9.8)$$

$$J = \frac{1}{4} 160 \cdot 9.8 = 40 \cdot 9.8 = \boxed{392 \text{ N}}$$

6/10



$$L = 5 \text{ m}$$

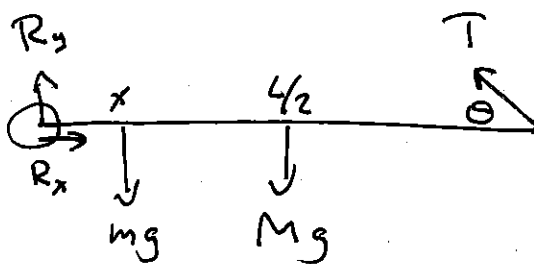
$$T_{\text{max}} = 10^4 \text{ N}$$

$$M = 1000 \text{ kg}$$

$$m = 100 \text{ kg}$$

$$\theta = 36^\circ$$

$$x_{\text{max}} = ?$$



Do torque at left
to remove reaction forces.

$$\sum \tau_{\text{left}}: -xmg - \frac{L}{2}Mg + LT \sin \theta = 0$$

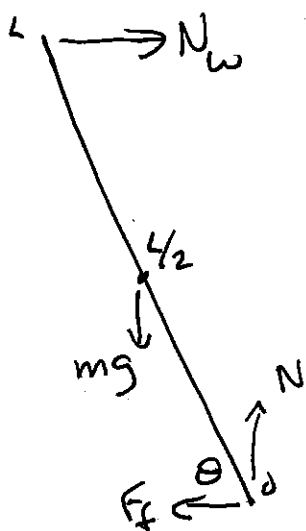
$$xmg = L \left(-\frac{Mg}{2} + T \sin \theta \right)$$

$$x = \frac{L}{mg} \left(T \sin \theta - \frac{Mg}{2} \right)$$

$$x_{\text{max}} = \frac{5}{100 \cdot 9.8} \left(10^4 \cdot 0.5 - \frac{1000 \cdot 9.8}{2} \right)$$

$$x = 0.005102 (5000 - 4900)$$

$$x_{\text{max}} = 0.5102 \text{ m}$$



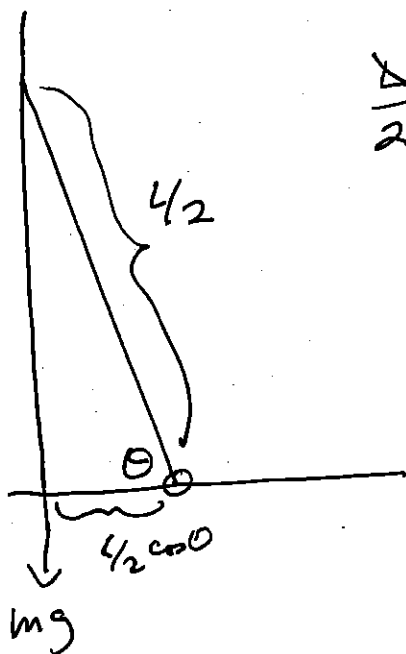
ladder leans at θ
 What is minimum θ
 for it not to slide?

7/10

$$\theta_{\min} \text{ means } F_{f \max} = \mu_s N$$

$$N_w = F_f = \mu_s mg$$

$$\sum \tau_{\text{base}}: \frac{L}{2} \cos \theta mg - L \sin \theta \mu_s mg = 0$$



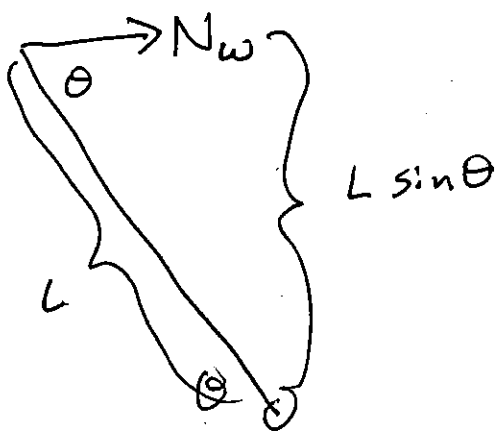
$$\frac{L}{2} \cos \theta mg = L \sin \theta \mu_s mg$$

$$\frac{\cos \theta}{2} = \mu_s \sin \theta \quad \frac{\sin}{\cos} \rightarrow \tan$$

$$\tan \theta = \frac{1}{2\mu_s}$$

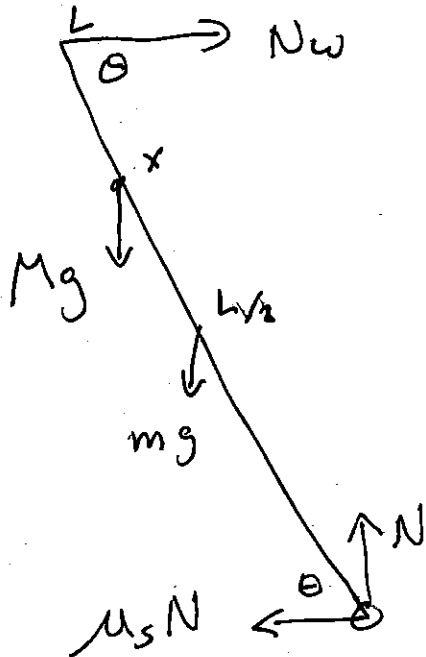
$$\theta_{\min} = \tan^{-1} \frac{1}{2\mu_s}$$

the minimum angle theta
 for the ladder to stand
not fall is $\tan^{-1} \frac{1}{2\mu_s}$



Leaning ladder with a man climbing it! 8/10
 how far x can he climb w/o falling?

$L = 10\text{m}$ $\theta = 65^\circ$ $\mu_s = 0.4$ $m = 10\text{kg}$ $M = 150\text{kg}$
 man plus load



$$\Sigma F_y: N - Mg - mg = 0$$

$$N = (M + m)g$$

$$\Sigma F_x: N_w - F_f = 0$$

$$N_w = F_f = \mu_s N = \mu_s g (M + m)$$

$$\Sigma \tau_{\text{base}}: \frac{L}{2} \cos \theta mg + x \cos \theta Mg$$

$$- N_w L \sin \theta$$

$$\frac{L}{2} \cos \theta mg + x \cos \theta Mg - L \sin \theta \mu_s g (M + m) = 0$$

$$x \cos \theta Mg = L \sin \theta (M + m) \mu_s g - \frac{L}{2} mg \cos \theta$$

cancel g divide by $\cos \theta$

$$Mx = L \tan \theta \mu_s (M + m) - \frac{L}{2} m$$

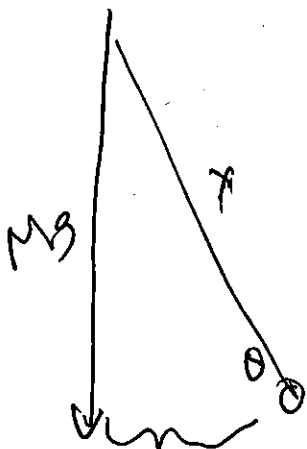
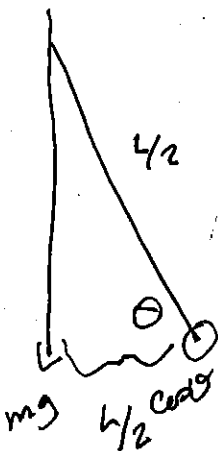
$$x = \frac{L}{M} \left(\mu_s (M + m) \tan \theta - \frac{m}{2} \right)$$

$$\tan 65^\circ = 2.1445$$

$$x = \frac{10}{150} \left(0.4 (150 + 10) 2.14 - \frac{10}{2} \right)$$

$$= 10 \frac{0.4 \cdot 2.14 \cdot 160 - 5}{150}$$

$$= 8.80\text{m}$$



What would get him to roof?

9/10

increase μ_s or θ

if θ $65^\circ \rightarrow 70^\circ$ $\tan 70^\circ = 2.75$

$$x = \frac{10}{150} (0.4 \cdot 160 \cdot 2.75 - 5) = 11.4 \text{ m}$$

μ_s $0.4 \rightarrow 0.6$, leave θ alone

$$x = \frac{10}{150} (0.6 \cdot 160 \cdot 2.14 - 5) = 13.4 \text{ m}$$

He can climb to $x=L$ what θ is minimum?

$$xM = \mu_s L (M+m) \tan \theta - \frac{mL}{2}$$

$$LM + L \frac{m}{2} = \mu_s L (M+m) \tan \theta$$

$$M + \frac{m}{2} = (M+m) \mu_s \tan \theta$$

$$\tan \theta = \frac{M + \frac{m}{2}}{(M+m) \mu_s} \Rightarrow$$

$$\theta_{\min} = \tan^{-1} \frac{150+5}{(150+10) \cdot 0.4}$$

$$\theta_{\min} = \tan^{-1} \frac{155}{0.4 \cdot 160}$$

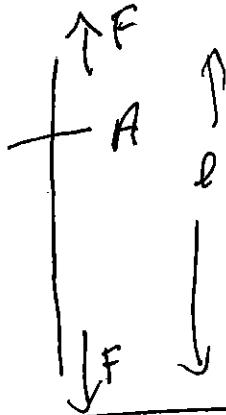
$$= \tan^{-1} 2.42 = \boxed{67.6^\circ}$$

STRESS/STRAIN

10/10

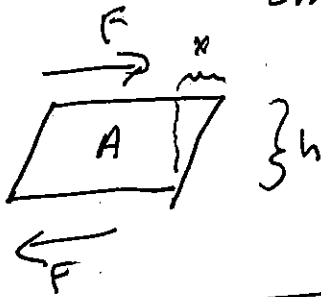
YOUNG'S MODULUS

$$Y = \frac{\text{tensile stress}}{\text{tensile strain}} = \frac{F/A}{\Delta l/l}$$



SHEAR

$$S = \frac{\text{shear stress}}{\text{shear strain}} = \frac{F/A}{\Delta x/h}$$



BULK

$$B = \frac{\text{volume stress}}{\text{volume strain}} = \frac{-\Delta F/A}{\Delta V/V_0} = \frac{-P}{\Delta V/V_0}$$

W

$$Y = 35 \cdot 10^{10} \text{ N/m}^2$$

$$S = 14 \cdot 10^{10}$$

$$B = 20 \cdot 10^{10}$$

Al

$$Y = 7 \cdot 10^{10}$$

$$S = 25 \cdot 10^{10}$$

$$B = 7 \cdot 10^{10}$$